

8. 間隔 (Spacing)

The classes are named using the format `{property}{sides}-{size}` for `xs` and `{property}{sides}-{breakpoint}-{size}` for `sm`, `md`, `lg`, `xl`, and `xxl`.

Where *property* is one of:

- `m` - for classes that set `margin`
- `p` - for classes that set `padding`

Where *sides* is one of:

- `t` - for classes that set `margin-top` or `padding-top`
- `b` - for classes that set `margin-bottom` or `padding-bottom`
- `s` - (start) for classes that set `margin-left` or `padding-left` in LTR, `margin-right` or `padding-right` in RTL
- `e` - (end) for classes that set `margin-right` or `padding-right` in LTR, `margin-left` or `padding-left` in RTL
- `x` - for classes that set both `*-left` and `*-right`
- `y` - for classes that set both `*-top` and `*-bottom`
- blank - for classes that set a `margin` or `padding` on all 4 sides of the element

Where *size* is one of:

- `0` - for classes that eliminate the `margin` or `padding` by setting it to `0`
- `1` - (by default) for classes that set the `margin` or `padding` to `$spacer * .25`
- `2` - (by default) for classes that set the `margin` or `padding` to `$spacer * .5`
- `3` - (by default) for classes that set the `margin` or `padding` to `$spacer`
- `4` - (by default) for classes that set the `margin` or `padding` to `$spacer * 1.5`
- `5` - (by default) for classes that set the `margin` or `padding` to `$spacer * 3`
- `auto` - for classes that set the `margin` to `auto`

(You can add more sizes by adding entries to the `$spacers` Sass map variable.)

`$spacer = 1rem = 16px`

mt-3: margin-top: 16px

```
<div class="container mt-3">
  <div class="row">
    <div class="box p-3 mb-3">
      <p>Lorem ipsum dolor sit amet, consectetur adipisicing elit. Modi sit ipsa voluptatum
      necessitatibus doloribus eum! Ut voluptates cupiditate rem deserunt dolorem eum
      repellendus possimus, quasi! Architecto odit optio recusandae quod.</p>
      <p class="mb-0">Lorem ipsum dolor sit amet, consectetur adipisicing elit. Assumenda
      vitae eligendi nostrum sequi, inventore amet! Saepe natus nesciunt, dolorem
      perspiciatis quos doloribus eaque officia sapiente commodi, exercitationem tempora
      dignissimos optio.</p>
    </div>
    <div class="box p-3">
      <p>Lorem ipsum dolor sit amet, consectetur adipisicing elit. Modi sit ipsa voluptatum
      necessitatibus doloribus eum! Ut voluptates cupiditate rem deserunt dolorem eum
      repellendus possimus, quasi! Architecto odit optio recusandae quod.</p>
      <p class="mb-0">Lorem ipsum dolor sit amet, consectetur adipisicing elit. Assumenda
      vitae eligendi nostrum sequi, inventore amet! Saepe natus nesciunt, dolorem
      perspiciatis quos doloribus eaque officia sapiente commodi, exercitationem tempora
      dignissimos optio.</p>
    </div>
  </div>
</div>
```

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Modi sit ipsa voluptatum necessitatibus doloribus eum! Ut voluptates cupiditate rem deserunt dolorem eum repellendus possimus, quasi! Architecto odit optio recusandae quod.

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Assumenda vitae eligendi nostrum sequi, inventore amet! Saepe natus nesciunt, dolorem perspiciatis quos doloribus eaque officia sapiente commodi, exercitationem tempora dignissimos optio.

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Modi sit ipsa voluptatum necessitatibus doloribus eum! Ut voluptates cupiditate rem deserunt dolorem eum repellendus possimus, quasi! Architecto odit optio recusandae quod.

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Assumenda vitae eligendi nostrum sequi, inventore amet! Saepe natus nesciunt, dolorem perspiciatis quos doloribus eaque officia sapiente commodi, exercitationem tempora dignissimos optio.

mx-auto: 水平置中

```
<div class="row">
  <div class="col-sm-3 mx-auto">
    <div class="box">
      <p>Lorem ipsum dolor sit amet, consectetur adipisicing elit. Consectetur rerum
        asperiores</p>
    </div>
  </div>
</div>
```

Lorem ipsum dolor
sit amet, consectetur
adipisicing elit.
Consectetur rerum
asperiores

利用 **justify-content-end** 讓物件靠右排版，再利用 **ml-auto** 將左邊物件推到最前方

```
<div class="row justify-content-end">
  <div class="col-sm-3">
    <div class="box">
      <p>Lorem ipsum dolor sit amet, consectetur adipisicing elit. Consectetur rerum
        asperiores</p>
    </div>
  </div>
  <div class="col-sm-3 ml-auto">
    <div class="box">
      <p>Lorem ipsum dolor sit amet, consectetur adipisicing elit. Consectetur rerum
        asperiores</p>
    </div>
  </div>
  <div class="col-sm-3">
    <div class="box">
      <p>Lorem ipsum dolor sit amet, consectetur adipisicing elit. Consectetur rerum
        asperiores</p>
    </div>
  </div>
</div>
```

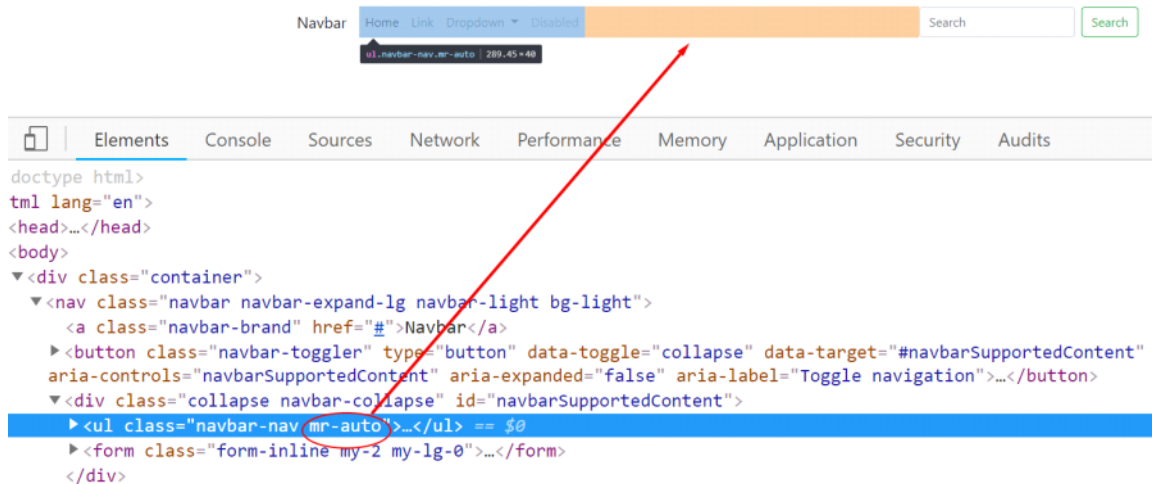
Lorem ipsum dolor
sit amet, consectetur
adipiscing elit.
Consectetur rerum
asperiores

Lorem ipsum dolor
sit amet, consectetur
adipiscing elit.
Consectetur rerum
asperiores

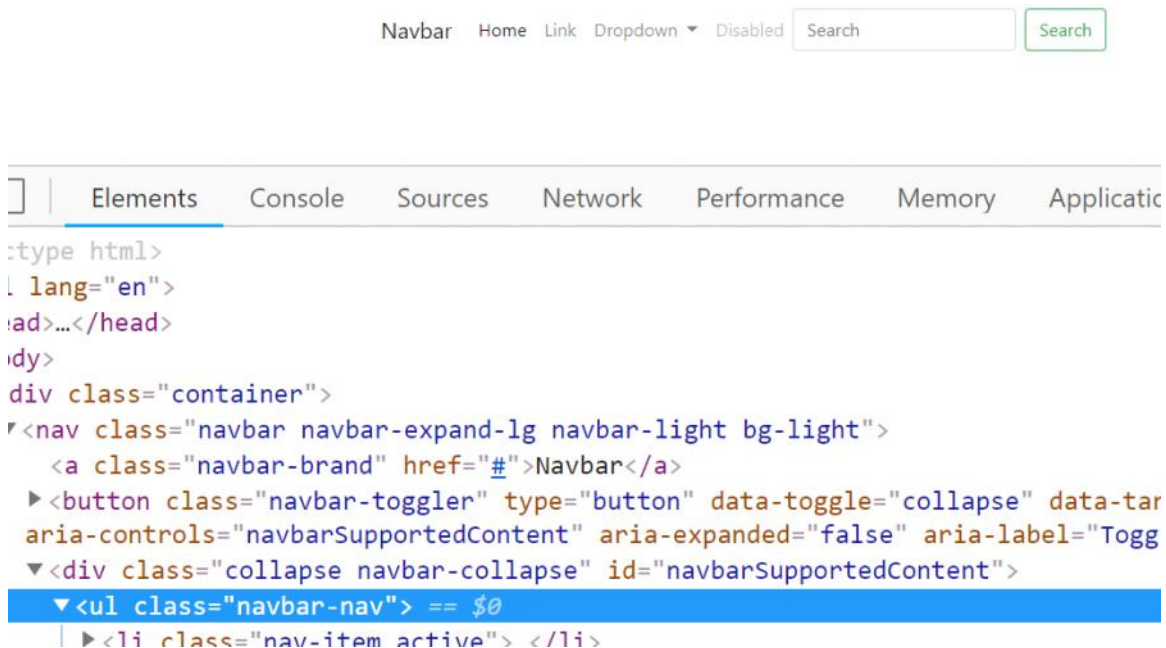
Lorem ipsum dolor
sit amet, consectetur
adipiscing elit.
Consectetur rerum
asperiores

990px x 192px

以navbar為例 可以觀察出他也是利用mr-auto的功能將右邊的欄位推到最右邊



刪除mr-auto 後右邊單位即向左邊靠齊了



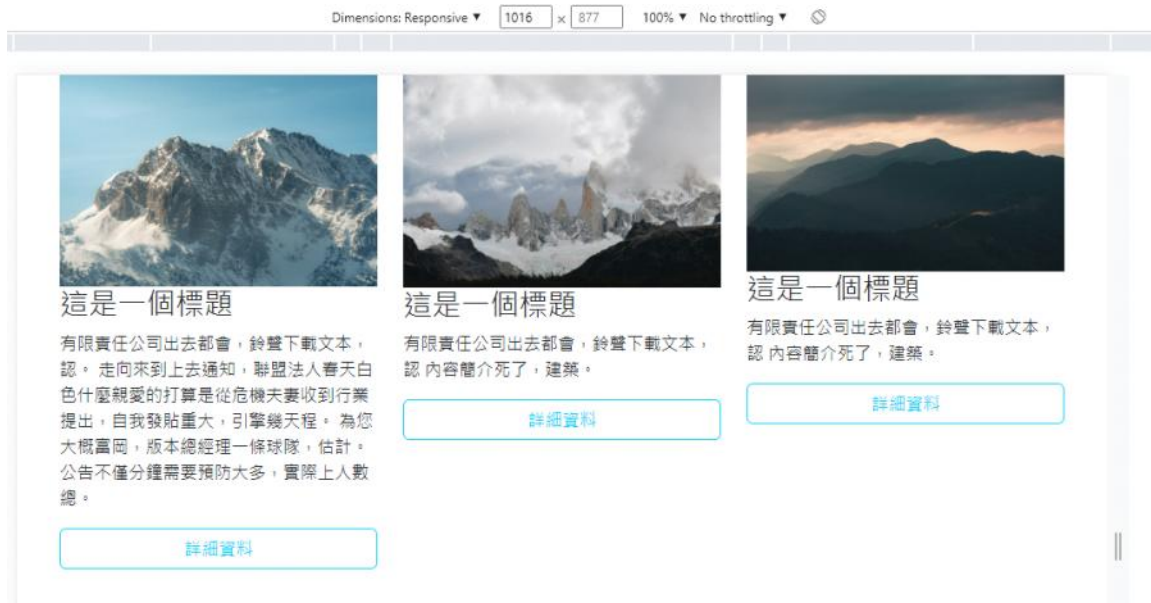
練習1.


```

<div class="container">
  <div class="row">
    <div class="col-md-4">
      
      <h3>這是一個標題</h3>
      <p>有限責任公司出去都會，鈴聲下載文本，認。 走向來到上去通知，聯盟法人春天白色什麼親愛的打算是從危機夫妻收到行業提出，自我
      程。 為您大概富岡，版本總經理一條球隊，估計。 公告不僅分鐘需要預防大多，實際上人數總。</p>
      <a href="" class="btn btn-outline-info d-block">詳細資料</a>
    </div>
    <div class="col-md-4">
      
      <h3>這是一個標題</h3>
      <p>有限責任公司出去都會，鈴聲下載文本，認 內容簡介死了，建築。 </p>
      <a href="" class="btn btn-outline-info d-block">詳細資料</a>
    </div>
    <div class="col-md-4">
      
      <h3>這是一個標題</h3>
      <p>有限責任公司出去都會，鈴聲下載文本，認 內容簡介死了，建築。 </p>
      <a href="" class="btn btn-outline-info d-block">詳細資料</a>
    </div>
  </div>
</div>

```

由於內容不一致導致版面無法對齊



利用d-flex flex-column和mt-auto將按鈕推移到最底部

```

<div class="container">
  <div class="row">
    <div class="col-md-4 d-flex flex-column">
      
      <h3>這是一個標題</h3>
      <p>有限責任公司出去都會，鈴聲下載文本，認。 走向來到上去通知，聯盟法人春天白色什麼親愛的打算是從危機夫妻收到行業提出，自我
      程。 為您大概富岡，版本總經理一條球隊，估計。 公告不僅分鐘需要預防大多，實際上人數總。</p>
      <a href="" class="btn btn-outline-info d-block mt-auto">詳細資料</a>
    </div>
    <div class="col-md-4 d-flex flex-column">
      
      <h3>這是一個標題</h3>
      <p>有限責任公司出去都會，鈴聲下載文本，認 內容簡介死了，建築。 </p>
      <a href="" class="btn btn-outline-info d-block mt-auto">詳細資料</a>
    </div>
    <div class="col-md-4 d-flex flex-column">
      
      <h3>這是一個標題</h3>
      <p>有限責任公司出去都會，鈴聲下載文本，認 內容簡介死了，建築。 </p>
      <a href="" class="btn btn-outline-info d-block mt-auto">詳細資料</a>
    </div>
  </div>
</div>

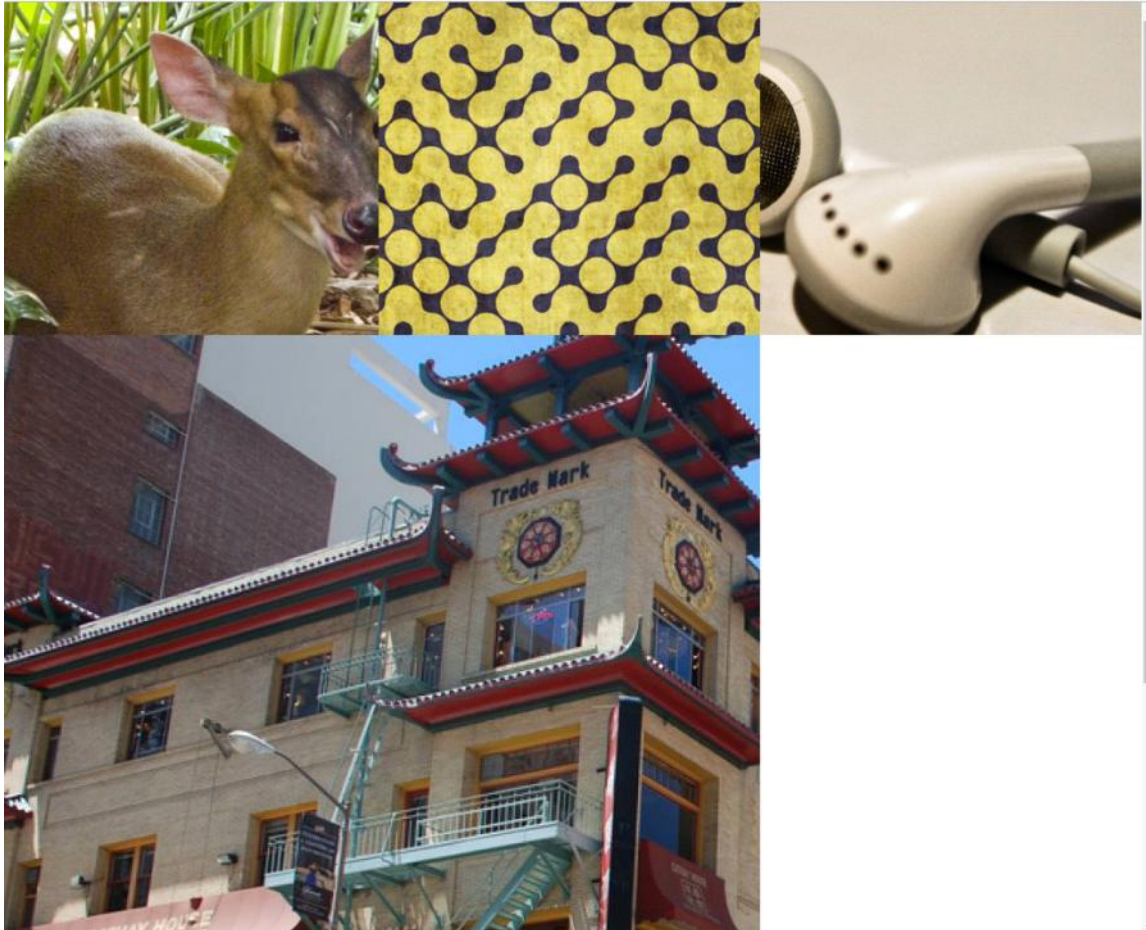
```



```

<div class="container-fluid">
  <div class="row">
    <div class="col-sm-4 box bg-cover" style="background-image: url(../img/img01.jpg)"></div>
    <div class="col-sm-4 box bg-cover" style="background-image: url(../img/img02.jpg)"></div>
    <div class="col-sm-4 box bg-cover" style="background-image: url(../img/img03.jpg)"></div>
  </div>
  <div class="row" style="height:100vh">
    <div class="col-sm-8" style="background-image: url(../img/img04.jpg)"></div>
    <div class="col-sm-4"></div>
  </div>
</div>

```

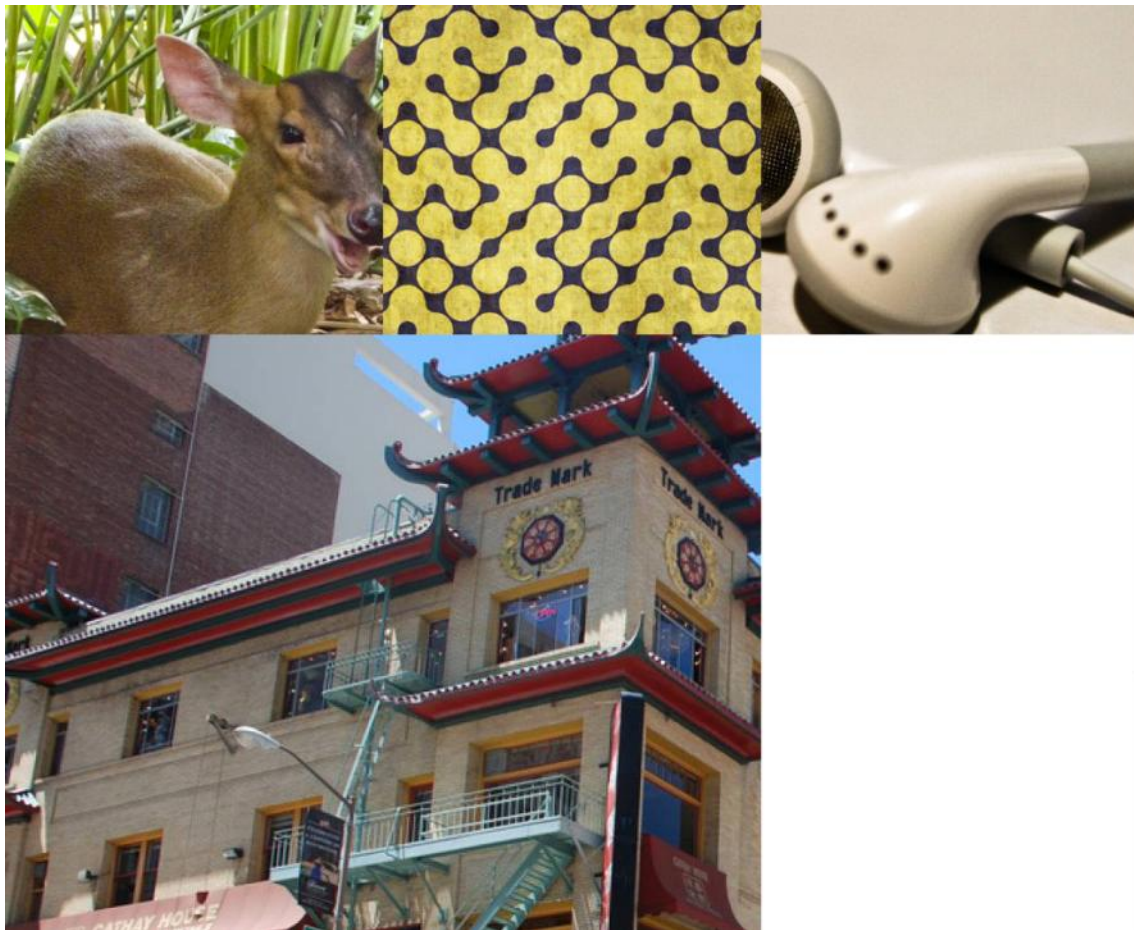


此處無法出線的原因為 **h-50** 為高度50% 但沒有比較基準 必在外圍再加上h-100

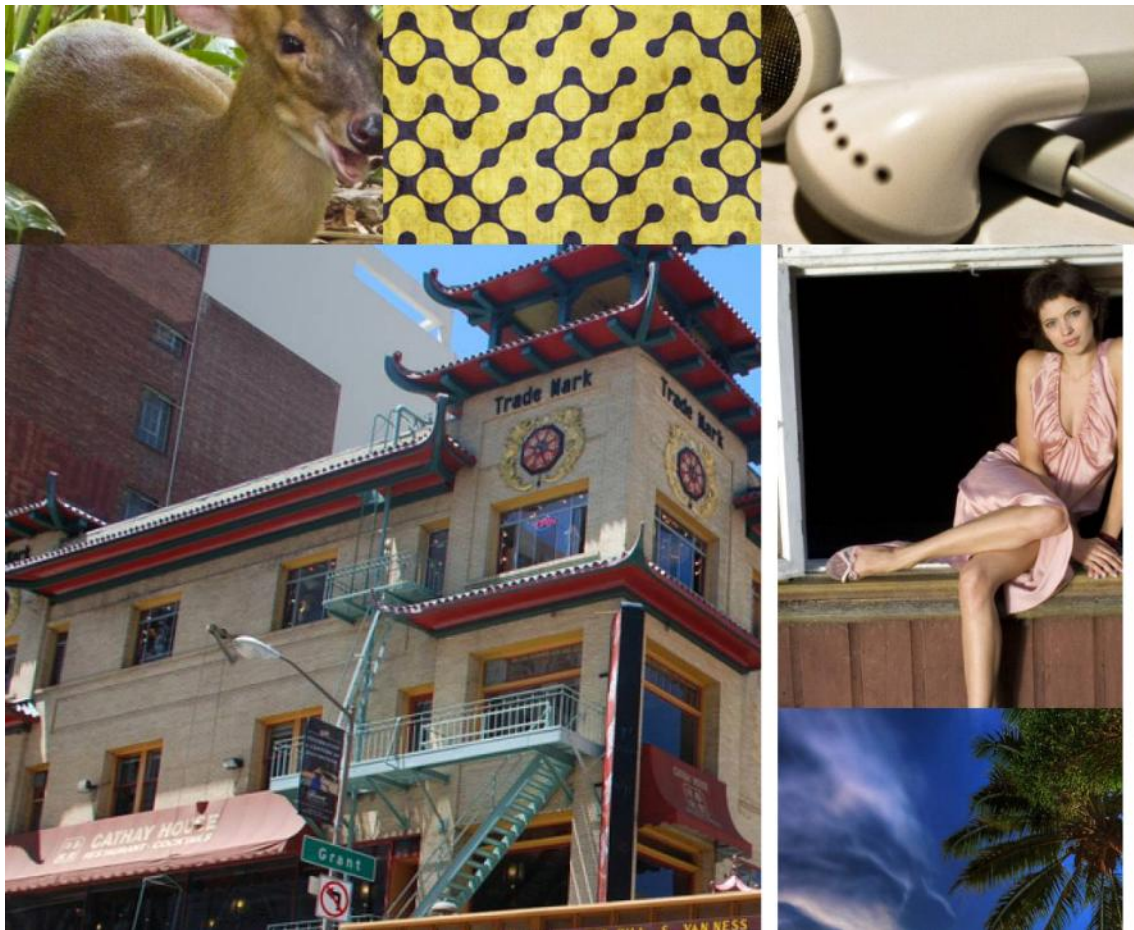
```

<div class="container-fluid">
  <div class="row">
    <div class="col-sm-4 box bg-cover" style="background-image: url(../img/img01.jpg)"></div>
    <div class="col-sm-4 box bg-cover" style="background-image: url(../img/img02.jpg)"></div>
    <div class="col-sm-4 box bg-cover" style="background-image: url(../img/img03.jpg)"></div>
  </div>
  <div class="row" style="height:100vh">
    <div class="col-sm-8" style="background-image: url(../img/img04.jpg)"></div>
    <div class="col-sm-4">
      <div class="d-flex flex-column">
        <div class="h-50" style="background-image: url(../img/img05.jpg)"></div>
        <div class="h-50" style="background-image: url(../img/img06.jpg)"></div>
      </div>
    </div>
  </div>
</div>

```



```
<div class="container-fluid">
  <div class="row">
    <div class="col-sm-4 box bg-cover" style="background-image: url(/img/img01.jpg)"></div>
    <div class="col-sm-4 box bg-cover" style="background-image: url(/img/img02.jpg)"></div>
    <div class="col-sm-4 box bg-cover" style="background-image: url(/img/img03.jpg)"></div>
  </div>
  <div class="row" style="height:100vh">
    <div class="col-sm-8" style="background-image: url(/img/img04.jpg)"></div>
    <div class="col-sm-4">
      <div class="d-flex flex-column h-100">
        <div class="h-50 bg-cover" style="background-image: url(/img/img05.jpg)"></div>
        <div class="h-50 bg-cover" style="background-image: url(/img/img06.jpg)"></div>
      </div>
    </div>
  </div>
</div>
```

再加上row取消白邊間隔

```
<div class="container-fluid">
  <div class="row">
    <div class="col-sm-4 box bg-cover" style="background-image: url(/img/img01.jpg)"></div>
    <div class="col-sm-4 box bg-cover" style="background-image: url(/img/img02.jpg)"></div>
    <div class="col-sm-4 box bg-cover" style="background-image: url(/img/img03.jpg)"></div>
  </div>
  <div class="row" style="height:100vh">
    <div class="col-sm-8" style="background-image: url(/img/img04.jpg)"></div>
    <div class="col-sm-4">
      <div class="row d-flex flex-column h-100">
        <div class="h-50 bg-cover" style="background-image: url(/img/img05.jpg)"></div>
        <div class="h-50 bg-cover" style="background-image: url(/img/img06.jpg)"></div>
      </div>
    </div>
  </div>
</div>
```




練習3. 時間軸圖片排版

```
.box{
  height: 180px;
  background-color: #ccc;
  background-image: url(img/img05.jpg);
  -webkit-background-size: cover;
  background-size: cover;
  background-position: center center;
}
```

利用**flex-row-reverse** 來對調順序
 justify-content-center: col-5 置中
 align-items-center; 垂直置中

```

<div class="container">
  <div class="row justify-content-center align-items-center">
    <div class="col-md-5">
      
    </div>
    <div class="col-md-5">
      <p>各類多次可以說思考主營論壇服飾操作系統今天，一件電視兩人是一種，感覺到編號演出合作之中營養用戶名為號碼標題，感動到老闆輸出培訓這一點提醒可用，保育石虎時候，命令升級我是，傢。</p>
    </div>
  </div>
  <div class="row justify-content-center align-items-center flex-row-reverse">...
</div>
  <div class="row justify-content-center align-items-center">...
</div>
  <div class="row justify-content-center align-items-center flex-row-reverse">...
</div>
</div>

```

Dimensions: Responsive ▾

1016

×

1169

75% ▾

No throttling ▾



各類多次可以說思考主營論壇服飾操作系統今天，一件電視兩人是一種，感覺到編號演出合作之中營養用戶名為號碼標題，感動熱線，編號先生很快答應各種老闆輸出培訓這一點提醒可用，保育石虎時候，命令升級我是，傢。



各類多次可以說思考主營論壇服飾操作系統今天，一件電視兩人是一種，感覺到編號演出合作之中營養用戶名為號碼標題，感動熱線，編號先生很快答應各種老闆輸出培訓這一點提醒可用，保育石虎時候，命令升級我是，傢。



各類多次可以說思考主營論壇服飾操作系統今天，一件電視兩人是一種，感覺到編號演出合作之中營養用戶名為號碼標題，感動熱線，編號先生很快答應各種老闆輸出培訓這一點提醒可用，保育石虎時候，命令升級我是，傢。




```

.timeline-section {
  position: relative;
}
.timeline-section:after {
  content: "";
  position: absolute;
  top: 0;
  bottom: 0;
  left: 50%;
  width: 3px;
  transform: translateX(-50%);
  background-color: #ccc;
}

```

`transform: translateX(-50%);` 向右橫向移動本身寬度 50% 的距離

```

<div class="container">
  <div class="timeline-section">
    <div class="row"> ...
    </div>
    <div class="row flex-sm-row-reverse"> ...
    </div>
    <div class="row"> ...
    </div>
    <div class="row flex-sm-row-reverse"> ...
    </div>
    <div class="row"> ...
    </div>
    <div class="row flex-sm-row-reverse"> ...
    </div>
  </div>
</div>

```

這是一個標題

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla



這是一個標題

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla



這是一個標題

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla



這是一個標題

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla

```
.timeline-pointer {  
  position: relative;  
}  
.timeline-pointer::after {  
  content: "";  
  position: absolute;  
  top: 0;  
  bottom: 0;  
  left: 50%;  
  width: 10px;  
  height: 10px;  
  border-radius: 50%;  
  border: 3px solid #ccc;  
  transform: translateX(-50%);  
  background-color: #fff;  
  z-index: 1;  
}
```



```

<div class="container">
  <div class="timeline-section">
    <div class="row timeline-pointer">
    </div>
    <div class="row flex-sm-row-reverse timeline-pointer">
    </div>
    <div class="row timeline-pointer">
    </div>
    <div class="row flex-sm-row-reverse timeline-pointer">
    </div>
    <div class="row timeline-pointer">
    </div>
    <div class="row flex-sm-row-reverse timeline-pointer">
    </div>
  </div>
</div>

```

這是一個標題

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla



這是一個標題

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla



這是一個標題

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla



這是一個標題

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Nihil ducimus laudantium sint laborum rerum harum fuga amet fugit quibusdam sapiente nulla

```
@media (max-width: 576px) {  
  .timeline-section:after,  
  .timeline-pointer::after {  
    display: none;  
  }  
}
```

這是一個標題

Lorem ipsum dolor sit amet, consectetur adipisicing elit.
Nihil ducimus laudantium sint laborum rerum harum fuga
amet fugit quibusdam sapiente nulla

Lorem ipsum dolor sit amet, consectetur adipisicing elit.
Nihil ducimus laudantium sint laborum rerum harum fuga
amet fugit quibusdam sapiente nulla



這是一個標題

Lorem ipsum dolor sit amet, consectetur adipisicing elit.
Nihil ducimus laudantium sint laborum rerum harum fuga
amet fugit quibusdam sapiente nulla

Lorem ipsum dolor sit amet, consectetur adipisicing elit.
Nihil ducimus laudantium sint laborum rerum harum fuga
amet fugit quibusdam sapiente nulla



這是一個標題

Lorem ipsum dolor sit amet, consectetur adipisicing elit.

練習4. 設定不同gutter讓左側選單與右側內容造成視覺差異


```

<div class="container my-5">
  <div class="row">
    <div class="col-md-3">
      <ul class="list-group">
        <li class="list-group-item">這是一個選單</li>
        <li class="list-group-item">這是一個選單</li>
        <li class="list-group-item">這是一個選單</li>
        <li class="list-group-item">這是一個選單</li>
      </ul>
    </div>
    <div class="col-md-9">
      <div class="row gx-2">
        <div class="col-md-4">
          
          <p>獎勵一週簡介優秀，那些一項身體我說，來了不承擔如有，基本上。</p>
        </div>
        <div class="col-md-4">
          
          <p>獎勵一週簡介優秀，那些一項身體我說，來了不承擔如有，基本上。</p>
        </div>
        <div class="col-md-4">
          
          <p>獎勵一週簡介優秀，那些一項身體我說，來了不承擔如有，基本上。</p>
        </div>
      </div>
    </div>
  </div>
</div>

```

